TOBILOLA OGUNBOWALE

Biological Safety and AI Systems | Microbiology | Machine Learning

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PROFESSIONAL SUMMARY

Biological scientist and ML engineer with an MS in Biomedical Informatics, an MS in Medical Microbiology, and 10 years of experience across clinical diagnostics, molecular biology, and biomedical research. I build ML and LLM-assisted tools for laboratory and healthcare environments with a focus on safety, oversight, auditability, and clear documentation in regulated workflows. Recent work includes a dual-use biological query screening prototype grounded in DURC policy, the Biological Weapons Convention, and Australia Group frameworks, alongside production lab informatics systems supporting data integrity, anomaly detection, and compliance reporting.

EDUCATION

Master of Science, Biomedical Informatics | University at Buffalo, SUNY | 2023
Coursework: Machine Learning, Deep Learning, Bioinformatics Algorithms, Health Data Analytics, Clinical Decision Support Systems

Master of Science, Medical Microbiology | University of Lagos | 2016
Focus: Clinical Microbiology, Molecular Diagnostics, Laboratory Quality Management

Bachelor of Science, Microbiology | Redeemer's University, Nigeria | 2012

TECHNICAL PROJECTS

Requiva Lab Management System [Production - In Active Use] github.com/tobilola/Requiva

- Production system currently in use at University at Buffalo. Built five ML models covering predictive inventory reordering, spending forecasts, anomaly detection, and vendor optimization to reduce manual overhead in a research lab setting.
- Developed under real operational constraints with NIH compliance requirements, audit-ready logging, and QC checks that had to hold up under daily use by non-technical lab staff.
- Evaluated models on precision and recall, iterated on thresholds, and built a Streamlit dashboard for monitoring and reporting.

Stack: *Python, Streamlit, Firebase, Scikit-learn, Pandas*

BioSafe Classifier: LLM Query Safety Screening System [Portfolio Project] github.com/tobilola/biosafe-classifier

- Built a personal project to explore biological query classification for dual-use risk. Designed a five-level risk taxonomy grounded in DURC policy, Australia Group guidelines, and the Biological Weapons Convention, and used it to label a hand-curated evaluation set.
- Extracted 36 features per query including topic category matching, intent escalation signals, and specificity indicators to help distinguish legitimate research questions from potential misuse attempts.
- Tracked false positives against legitimate research queries and tuned thresholds by risk level to reduce unnecessary refusals. Ran stress tests using adversarial prompt variants including paraphrase, obfuscation, and benign framing to identify bypass patterns, and documented failure modes and mitigation ideas.
- Trained a Random Forest and Gradient Boosting ensemble with stratified cross-validation. Built a Streamlit dashboard for live classification, batch analysis, and performance visualization. Documented known limitations including keyword brittleness and single-query context, with a roadmap for BioBERT embeddings.

Stack: *Python, Scikit-learn, Streamlit, Pandas, NumPy*

Medical AI Agent with Retrieval-Augmented Generation [Portfolio Project] github.com/tobilola/medical-ai-agent

- A personal project exploring clinical decision support using LLMs with tool use and retrieval. Emphasized source-grounding, structured outputs, and error handling as mechanisms for reducing hallucinations and supporting human review.
- Integrated PubMed search, drug interaction checks, clinical calculators, and patient data querying into a unified agent framework. Built a RAG system using ChromaDB indexed on clinical guidelines and protocols, validated on a small pilot set of 100 queries, and iterated based on output review.
- Deployed via FastAPI and Streamlit with structured output parsing and error handling for safe, reliable responses.

Stack: *LangChain, GPT-4, ChromaDB, FastAPI, PostgreSQL, Python*

Regulatory Compliance Tracker with NLP [Portfolio Project] github.com/tobilola/compliance-tracker

- A portfolio project building automated compliance monitoring using OCR and NLP to classify regulatory documents and detect compliance gaps. Trained and evaluated multiple models for document classification and gap detection, reporting precision and recall and reviewing failures to reduce high-impact misclassifications.
- Implemented entity extraction using spaCy and Transformers to identify regulatory requirements across document types. Designed a real-time dashboard for compliance risk visualization and audit-ready reporting. Models achieved 94.5% accuracy on the held-out evaluation set.

Stack: *Tesseract OCR, Scikit-learn, spaCy, Transformers, React*

LIMS-ELN Integration Platform [Portfolio Project] github.com/tobilola/LIMS-ELN_Integration

- A portfolio project building middleware for bidirectional data synchronization between laboratory information management and electronic lab notebook systems. Included audit-ready logging, integrity checks, and anomaly signals designed to support review and incident investigation.
- Implemented anomaly detection, NER-based metadata tagging, and automated compliance scoring for laboratory data integrity. Designed asynchronous pipelines using Redis queues to handle 1000 samples per day with full audit logging.

Stack: *FastAPI, PostgreSQL, MongoDB, Redis, spaCy, Docker*

PneumoniaNet AI: Deep Learning Diagnostic System [Portfolio Project] github.com/tobilola/pneumonianet-ai

- A portfolio project exploring medical image analysis using deep learning. Designed a custom CNN combining ResNet50 with spatial attention mechanisms for pneumonia detection from chest X-rays.
- Built a full training pipeline with AdamW optimization, learning rate scheduling, gradient clipping, and attention visualization. Delivered a production inference system with batch processing and interpretable model outputs intended for clinical review.

Stack: *PyTorch, torchvision, ResNet50, React, Python*

PROFESSIONAL EXPERIENCE

Research Systems Operations Manager

University at Buffalo, Department of Medicine | Buffalo, NY | *2024 to Present*

- Design AI-powered automation for inventory tracking, equipment scheduling, and workflow optimization across a $2M+ research portfolio.
- Deploy predictive analytics and automated QC procedures, reducing operational expenses by 60% and manual data errors by 85%.
- Lead cross-functional teams integrating laboratory information systems with clinical and research workflows under HIPAA, IRB, and IACUC compliance.
- Build real-time analytics dashboards improving resource forecasting accuracy by 20%.

Graduate Research Assistant, Bioinformatics and Laboratory Informatics

SUNY Center for Excellence in Bioinformatics and Life Sciences | Buffalo, NY | *Nov 2022 to Dec 2023*

- Built Snakemake and Nextflow pipelines for RNA-seq, whole-genome sequencing, and single-cell analyses. Processed 100 GB+ of sequencing data with automated QC and reporting.
- Implemented genomic QC pipelines using Seurat, Cell Ranger, and GATK for variant calling and quality assessment.
- Configured LIMS workflows for biospecimen tracking and developed Python scripts for preprocessing and reporting, reducing analysis time by 40%.

Digital Transformation Lead, Laboratory Operations

Afri-Global Medicare Limited | Lagos, Nigeria | *Jun 2020 to Aug 2021*

- Led enterprise-wide LIS implementation during COVID-19, supporting multi-site operations at high volume and increasing testing capacity by approximately 300% while maintaining quality controls.
- Achieved 95% user adoption across multi-site diagnostic operations and reduced turnaround time by 35% through workflow automation.
- Managed quality systems and biosafety compliance for high-complexity clinical laboratory operations.

Healthcare Technology Analyst

Field Approach Consult | Lagos, Nigeria | *Feb 2017 to May 2020*

- Managed LIMS and clinical data systems for a 500+ participant epidemiological study. Supported stakeholder reporting and regulatory-quality documentation for external partners.
- Coordinated cross-functional teams to implement quality systems, SOPs, and laboratory technology platforms across multiple sites.

Laboratory Coordinator and Analyst

University of Lagos, Central Research Laboratory | Lagos, Nigeria | *2014 to 2017*

- Performed molecular diagnostics including NGS workflows, PCR, RT-PCR, ELISA, and specimen processing.
- Implemented Lean Six Sigma quality systems and led equipment validation, calibration, and preventive maintenance programs.

TECHNICAL SKILLS

Programming: Python, R, SQL, JavaScript, Bash

ML and AI: PyTorch, Scikit-learn, XGBoost, LangChain, GPT-4 API, ChromaDB, spaCy, Transformers, RAG systems, anomaly detection, time-series forecasting (ARIMA, Prophet)

Bioinformatics: Snakemake, Nextflow, Seurat, Cell Ranger, GATK, BWA, SAMtools, STAR, DESeq2, HPC/Slurm, genomic data analysis

Laboratory Informatics: LIMS/LIS (EPIC Beaker, Cerner), instrument integration (HL7, ASTM), ELN, QC systems, lab automation

Regulatory Compliance: CLIA, HIPAA, GLP, IRB, IACUC, HL7 FHIR

Biosecurity and AI Safety: Dual-use research of concern (DURC) policy, Australia Group guidelines, Biological Weapons Convention, select agent regulations, risk taxonomy design, LLM safety evaluation

Engineering: FastAPI, Flask, Docker, PostgreSQL, MongoDB, Redis, Firebase, React, Git, GitHub Actions